2022

YEAR 11
YEARLY
EXAMINATION

Mathematics Standard 2

General Instructions	• Reading time – 5 minutes • Working time – 2 hours • Write using black pen • Write your Student ID below and on page 5 • NESA approved calculators may be used • A reference sheet is provided at the back of this paper • For questions in Section II, show relevant mathematical reasoning and/or calculations
Total marks: 80	**Section I — 10 marks** (pages 2–4) • Attempt Questions 1–10 • Allow about 20 minutes for this section **Section II** — **70 marks** (pages 6–19) • Attempt Questions 11–31 • Allow about 1 hour 40 minutes for this section

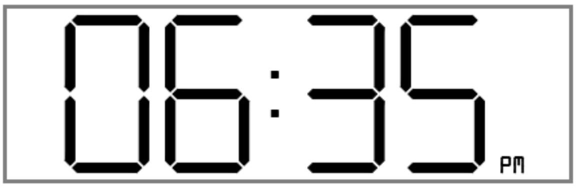
STUDENT ID: ______________________________

Section I

10 marks
Attempt Questions 1–10
Allow about 20 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

1 A clock shows the following time.



What is the time shown in 24-hour time?

A. 06:35

B. 16:35

C. 18:35

D. 26:35

2 Temperature, in degrees Celsius, is an example of which of the following classifications?

A. Numerical and continuous

B. Numerical and discrete

C. Categorical and nominal

D. Categorical and ordinal

3 Which of the following consists of three significant figures?

A. 8.470

B. 1.43×10^5

C. 2501

D. 0.67

4 Sasha invests \$8000 for nine months in a term deposit receiving simple interest at 6% p.a.

What is the interest earned on this investment?

A. \$360

B. \$1250

C. \$4320

D. \$8360

5 Consider the scores 12, 13, 15, 20, 21, 23, 36 and 40.

What is the mean?

A. 12

B. 20.5

C. 22.5

D. 40

6 What is the solution to the equation $3x - 14 = 13$?

A. $x = 1$

B. $x = 5$

C. $x = 9$

D. $x = 12$

7 Blood alcohol content (BAC) is calculated using the following formula:

$$BAC_{Male} = \frac{10N - 7.5H}{6.8M}$$

where N is the number of standard drinks consumed, H is the number of hours of drinking, and M is the person's weight in kilograms.

A glass of wine contains 1.2 standard drinks.

John weight 73 kg and consumes three glasses of wine between 2 pm and 4 pm.

What is John's BAC?

A. 0.0001

B. 0.0121

C. 0.0302

D. 0.0423

8 A pizza contains 2432 kilocalories. The pizza is cut into eight equal slices.

Mary eats two slices of pizza.

What amount of energy has Mary consumed, in kilojoules? (1 kilocalorie = 4.184 kJ).

A. 145.32 kJ

B. 581.26 kJ

C. 1162.52 kJ

D. 2543.87 kJ

9 Tom works from 9:00 am until 4:00 pm every Monday to Friday. He also works from 10 am until 1 pm on Saturday. He is paid $21.30 an hour on weekdays and time-and-a-half on weekends.

How much does Tom earn in a week?

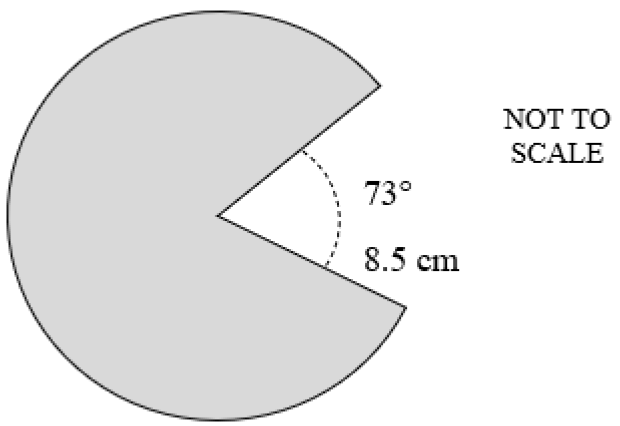
A. $745.50

B. $841.35

C. $975.25

D. $2130

10 The shaded shape is a sector.



What is its perimeter?

A. 59.58 cm

B. 62.69 cm

C. 105.72 cm

D. 226.98 cm

End of Section I

Student ID: ______________________

2022 **YEAR 11 YEARLY EXAMINATION**

Mathematics Standard 2
Section II Answer Booklet

70 marks
Attempt Questions 11–31
Allow about 1 hour 40 minutes for this part

Instructions

- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
- Your responses should include relevant mathematical reasoning and/or calculations
- Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering.
- Write your Student ID above

Please turn over

Question 11 (1 mark)

A food factory uses a sampling system to ensure quality. The take every twentieth can from a production line and check it for faults. **1**

What type of sampling is being used?

Question 12 (1 mark)

Jemima is catching a bus from Victoria cross to Central station. **1**

This is the timetable she would use.

Monday to Friday	♿	♿	♿	♿	♿	♿
Campbell St before Reserve Rd, Artarmon	08:50	09:00	09:15	09:30	09:45	10:00
St Leonards Station	08:57	09:06	09:21	09:36	09:51	10:06
Pacific Hwy after Albany St, St Leonards	08:58	09:07	09:22	09:37	09:52	10:07
Pacific Hwy after Falcon St, Crows Nest	09:01	09:09	09:24	09:39	09:54	10:09
Victoria Cross, Miller St, North Sydney	09:07	09:15	09:30	09:45	10:00	10:15
Wynyard Station	09:23	09:31	09:39	09:54	10:09	10:24
Town Hall Station	09:29	09:36	09:44	09:59	10:14	10:29
Central Station	09:41	09:46	09:54	10:09	10:24	10:39
Bourke St at Potter St, Waterloo	09:51	09:56	10:04	10:19	10:34	10:49
Gunyama Park Aquatic and Recreation Centre, Joynton Ave, Zetland	09:57	10:02	10:10	10:25	10:40	10:55

What is the latest Jemima can catch the bus from Victoria Cross to arrive at Central Station before 10:30 am?

Question 13 (1 mark)

The Kent family went on a road trip in the family car. **1**

Calculate the amount of petrol used by the car if the trip was 570 km and the car's petrol consumption was 6L/100 km.

Question 14 (2 marks)

A company buys a helicopter for \$12 659 and sells it three years later for \$10 205. **2**

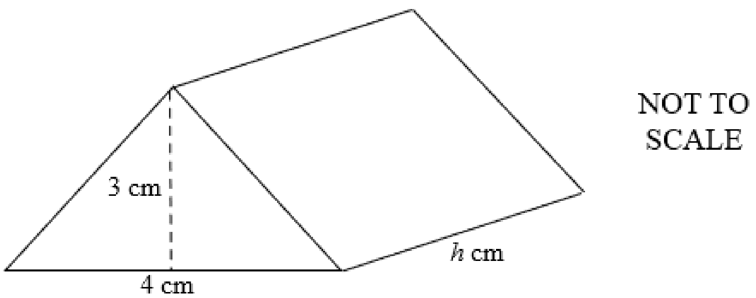
Calculate the loss as a percentage of the purchase price to one decimal place.

Question 15 (2 marks)

Rearrange the formula $R = \sqrt{\frac{ax}{b}}$ to make x the subject. **2**

Question 16 (2 marks)

The total volume of the prism shown is 120 cm^3. **2**



What is the value of h in the prism?

Question 17 (3 marks)

One Australian dollar (AUD) is the equivalent of 0.69 US dollars (USD).

(a) If Penelope wants to convert \$250 USD to AUD, how much money would she have in AUD? **1**

(b) One AUD is equivalent to 0.65 euro (EUR). **2**

Calculate how many USD are equivalent to 900 EUR.

Question 18 (2 marks)

Adelaide and Malaysia have Coordinated Universal Time of UTC +9.5 and UTC +8 respectively. **2**

Walter flew from Adelaide at 3:40 pm on Tuesday to travel to Malaysia. The flight took nine hours.

At what time and day did Walter arrive at Malaysia?

Question 19 (2 marks)

A brand of medicine has an adult dosage of 30 mL. **2**

Following the dosage carefully, Cecilia is given 24 mL when ill.

Using Clark's formula, calculate Cecilia's weight.

Clark's Formula:

$$Dosage = \frac{weight\ in\ kg\ \times\ adult\ dosage}{70}$$

Question 20 (3 marks)

The table shows the number of kilojoules burned per kilogram of body weight per hour for various activities. **3**

Activity	kJ/kg/h
Aerobics	12.3
Cycling	18.0
Running	58.7
Swimming	34.1
Yoga	8.4

Stuart has a body weight of 68 kg and spends two and a half hours exercising each day. One day he spent 30 minutes running, 50 minutes cycling and the remaining time swimming.

How many kilojoules did Stuart burn during this day, correct to the nearest integer?

Question 21 (3 marks)

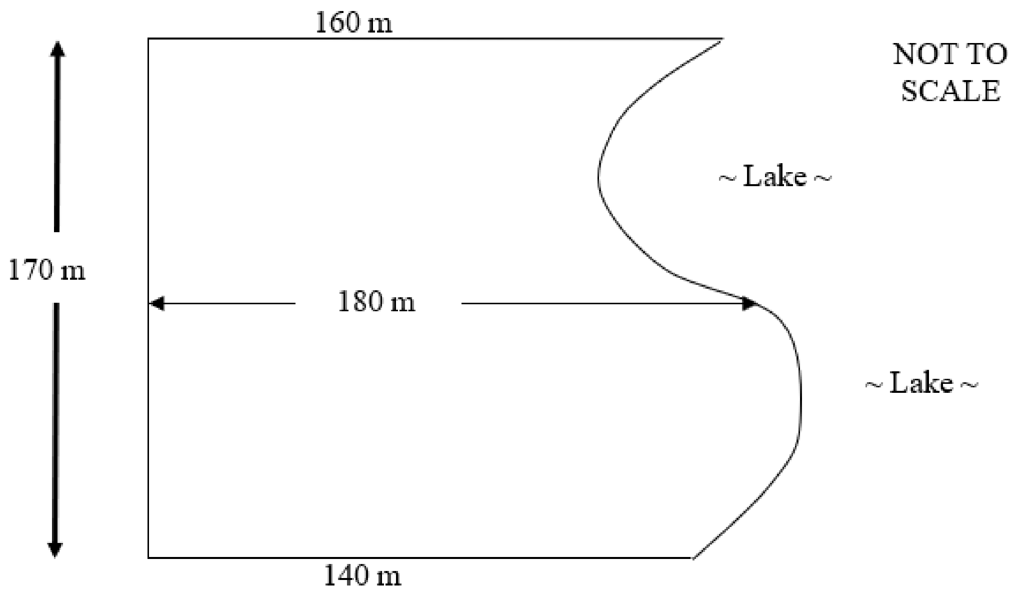
Eric is a doctor and earns $18 700 per month. Eric will be taking four weeks' annual leave. During this time, he will receive 17.5% annual leave loading on four weeks' pay in addition to his regular pay.

(a) Show that Eric's weekly wage is $4315.38. (Assume 52 weeks in 1 year) **1**

(b) Calculate Eric's total pay during his four weeks' annual leave. **2**

Question 22 (2 marks)

A golf course consists of an area of land surrounded by a lake as shown in the diagram. **2**



Use two applications of trapezoidal rule to calculate the area of the golf course.

Question 23 (3 marks)

Roger is driving on a busy road at a speed of 60 km/h when he must brake suddenly to avoid collision. He has a reaction time of 1.5 seconds. **3**

It is known that the braking distance of a car, in metres, is given by the formula $d = 0.01v^2$, where v is the velocity of the car in kilometres per hour.

Calculate Roger's **total stopping distance** using the following formula:

Total stopping distance = {reaction time distance} + {braking distance}

Question 24 (5 marks)

Chloe is hiring a function centre for a work event.

The cost of hiring the function centre is given by the equation $C = 3000 + 45n$ where C is the cost in dollars and n is the number of people that attend the function.

(a) How much would Chloe have to pay if she thinks 75 people will attend the function? **1**

(b) Chloe ends up needing to pay $8535. **2**

How many people attended the function?

(c) Sketch the linear relationship between cost (C) and number of people attending (n) on the grid below. **2**

Question 25 (6 marks)

An electric company charges per kWh based on the following rates:

$0.34 per kWh during peak time.
$0.16 per kWh during off-peak time.

Peak time is 7 am to 9 am and 5 pm to 8 pm on weekdays.
Off-peak is all other times.

Barry owns a refrigerator that runs 24 hours a day and consumes 2.88 kWh per day.

(a) How much energy does Barry's refrigerator use during an hour? **1**

(b) How much energy does the refrigerator use during peak time over **four weeks**? **1**

(c) How much energy does the refrigerator use during off peak time over **four weeks**? **2**

(d) Calculate the total cost to run Barry's refrigerator over **four weeks**. **2**

Question 26 (9 marks)

The following stem-and-leaf plot shows the scores achieved by a netballer in her last ten rounds.

Stem	Leaf
1	1 2 2 3
2	2 9
3	1 3 4
4	9

(a) Determine the median from the scores. **1**

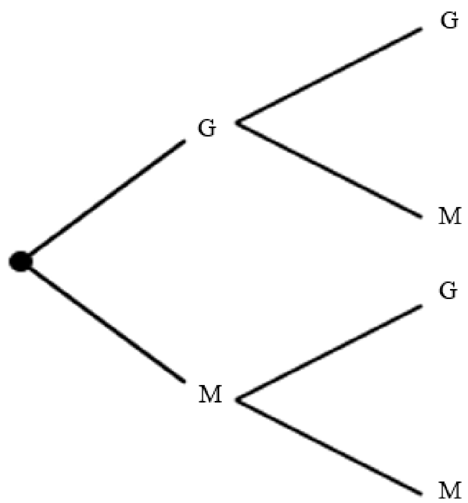
(b) Calculate the interquartile range. **2**

(c) Determine if there are any outliers in the dataset. Justify your answer with calculations. **3**

(d) Create a box and whisker plot to represent the data. **3**

Question 27 (5 marks)

Daniel is a goal kicker for a local rugby team. He has a 60% chance of kicking a goal.



(a) Complete the following tree diagram and list the possible outcomes when Daniel kicks twice. **2**

(b) Daniel kicks twice.

Determine the probability of Daniel scoring a goal and missing in any order, as a decimal. **2**

(c) How many goals would you expect Daniel to kick out of the twenty he takes? **1**

Question 28 (5 marks)

Jamal buys a new car for \$32 500. In addition to the sales price, he must pay 3% stamp duty.

Each month he must also pay CTP insurance of \$31.50 and every six months he must pay registration of \$212.

(a) What is the total cost of Jamal's car in the first year? **3**

(b) Jamal's car depreciates at a rate of 8% per annum of the initial cost.

Using the straight-line depreciation method, calculate the value of his vehicle after four years. **2**

Question 29 (3 marks)

The table shows the income tax rates for the 2021–2022 financial year. **3**

Taxable income ($)	*Tax payable*
$0 – $18 200	Nil
$18 201 – $45 000	19 cents for each $1 over $18 200
$45 001 – $120 000	$5092 plus 32.5 cents for each $1 over $45 000
$120 001 – $180 000	$29 467 plus 37 cents for each $1 over $120 000
$180 001 and over	$51 667 plus 45 cents for each $1 over $180 000

Sally earned $72 816 in the 2021–2022 financial year, working in an office for a bank.

She spends $520 a year on a company work uniform, $212 on office supplies and $6325 in petrol for her family-used car over the year.

Ignoring the Medicare levy, calculate Sally's tax payable for the 2021-2022 financial year.

Question 30 (6 marks)

A sign has dimensions of 29 cm x 38 cm, with each side measured to the nearest centimetre.

(a) What is the precision of the measurement? **1**

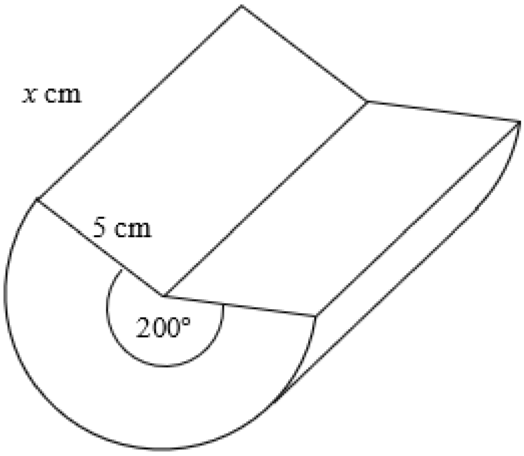
(b) What is the limit of accuracy for this measurement? **1**

(c) Calculate the percentage error of the shorter side correct to one decimal place. **1**

(d) Between what upper and lower limits does the perimeter of the sign lie? **3**

Question 31 (4 marks) **4**

A wooden log is in the shape of a cylinder. A part of this log has been removed and the remaining solid is shown below.



The surface area of the remaining log is 750 cm².

Find the value of *x* to the nearest millimetre.

End of paper

Section II extra writing space

If you use this space, clearly indicate which question you are answering.

Section II extra writing space

If you use this space, clearly indicate which question you are answering.

2022 YEAR 11 YEARLY EXAMINATION

Mathematics Standard 1
Mathematics Standard 2

REFERENCE SHEET

Measurement

Limits of accuracy

Absolute error = $\frac{1}{2} \times$ precision

Upper bound = measurement + absolute error

Lower bound = measurement — absolute error

Length

$$l = \frac{\theta}{360} 2\pi r$$

Area

$$A = \frac{\theta}{360} \pi r^2$$

$$A = \frac{h}{2}(a + b)$$

$$A \approx \frac{h}{2}(d_f + d_l)$$

Surface area

$$A = 2\pi r^2 + 2\pi rh$$

$$A = 4\pi r^2$$

Volume

$$V = \frac{1}{3}Ah$$

$$V = \frac{4}{3}\pi r^3$$

Trigonometry

$$\sin A = \frac{\text{opp}}{\text{hyp}}, \ \cos A = \frac{\text{adj}}{\text{hyp}}, \ \tan A = \frac{\text{opp}}{\text{adj}}$$

$$\text{A} = \frac{1}{2}ab\sin C$$

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab\cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

Financial Mathematics

$$FV = PV\,(1 + r)^n$$

Straight-line method of depreciation

$$S = V_o - Dn$$

Declining-balance method of depreciation

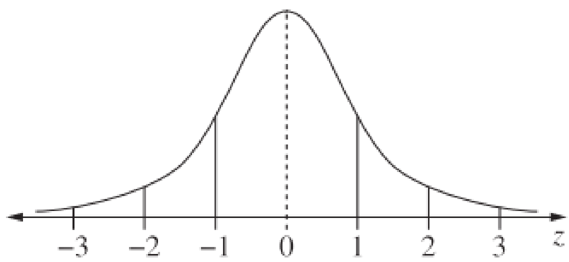
$$S = V_0\,(1 - r)^n$$

Statistical Analysis

An outlier is a score
less than $Q_1 - 1.5 \times IQR$
or
more than $Q_3 + 1.5 \times IQR$

$$z = \frac{x - \bar{x}}{s} \qquad z = \frac{x - \mu}{\sigma}$$

Normal distribution



- approximately 68% of scores have z-scores between -1 and 1
- approximately 95% of score have z-scores between -2 and 2
- approximately 99.7% of score have z-scores between -3 and 3

STUDENT ID ________________________

2022 Mathematics Standard 2 Year 11 Examination

Section I Answer Sheet

10 marks
Attempt Questions 1–10
Allow about 20 minutes for this section

Select the alternative A, B, C or D that best answers the question. Fill in the response circle completely.

1	A ○	B ○	C ○	D ○
2	A ○	B ○	C ○	D ○
3	A ○	B ○	C ○	D ○
4	A ○	B ○	C ○	D ○
5	A ○	B ○	C ○	D ○
6	A ○	B ○	C ○	D ○
7	A ○	B ○	C ○	D ○
8	A ○	B ○	C ○	D ○
9	A ○	B ○	C ○	D ○
10	A ○	B ○	C ○	D ○

2022 Mathematics Standard 2 Year 11
Marking Guidelines

Section I

Multiple-choice Answer Key

Question	Answer
1	C
2	A
3	B
4	A
5	C
6	C
7	D
8	D
9	B
10	A

Section II
Note: An incorrect answer in a previous part will not necessarily preclude students from achieving full marks in a later part. Answers here are based on correct prior part answers. Marking will need to adapt to pursue correct method with the use of incorrect prior parts.

Question 11 (1 marks)

Criteria	Marks
• Provides correct answer	1

Sample answer:
Systematic sampling

Question 12 (1 mark)

Criteria	Marks
• Provides correct answer	1

Sample answer:
Latest bus would arrive at Central Station at 10:24 am. This bus leaves Victoria Cross at 10 am.
∴10 am is the latest Jemima could leave Victoria Cross to arrive at Central Station.

Question 13 (1 mark)

Criteria	Marks
• Provides correct answer	1

Sample answer:
(570/100) x 6 = 34.2L

Question 14 (2 marks)

Criteria	Marks
• Provides correct answer	2
• Makes progress towards answer	1

Sample answer:
12659 – 10205 = 2454 loss

$$\frac{2454}{12659} \times 100 = 19.4\%$$

Question 15 (2 marks)

Criteria	Marks
• Provides correct answer	2
• Makes progress towards answer	1

Sample answer:

$R^2 = \frac{ax}{b}$

$R^2 b = ax$

$x = \frac{R^2 b}{a}$

Question 16 (2 marks)

Criteria	Marks
• Provides correct answer	2
• Makes progress towards answer	1

Sample answer:

V=Ah

$120 = \frac{(3 \times 4)}{2} \times h$

h = 20 cm

Question 17 (3 marks)

a

Criteria	Marks
• Provides correct answer	1

Sample answer:

0.69×250

$= \$172.50$

b

Criteria	Marks
• Provides correct answer	2
• Makes progress towards answer	1

Sample answer:

A = 900 x 0.65

A = $585

USD = $\frac{585}{0.69}$

= $847.83

Question 18 (2 marks)

Criteria	Marks
• Provides correct answer	2
• Makes progress towards answer	1

Sample answer:
1.5-hour difference
3:40 pm Adelaide is equivalent to 2:10 pm Malaysian time
9-hour flight
∴ 2:10 + 9 = 11:10 pm Tuesday

Question 19 (2 marks)

Criteria	Marks
• Provides correct answer	2
• Makes progress towards answer	1

Sample answer:

$$24 = \frac{weight\ in\ kg \times 30}{70}$$

$$Weight\ in\ kg \times 30 = 1\ 680$$

$$Weight = 51.94$$

$$= 56\ \text{kg}$$

Question 20 (3 marks)

Criteria	Marks
• Provides correct answer	3
• Makes progress towards answer	2
• Provides some relevant information	1

Sample answer:
2.5 hours = 150 minutes
150 – 80 = 70 minutes remaining

$$\left(\frac{30}{60} \times 58.7 \times 68\right) + \left(\frac{50}{60} \times 18.0 \times 68\right) + \left(\frac{70}{60} \times 34.1 \times 68\right)$$

= 5721 kJ

Question 21 (3 marks)

a

Criteria	Marks
• Provides correct answer	1

Sample answer:

$(18700 \times 12) \div 52$
= \$4315.38

b

Criteria	Marks
• Provides correct answer	2
• Makes progress towards answer	1

Sample answer:

$(4315.38 \times 4) + (4315.38 \times 4 \times 0.175)$
= \$20 282.29

Question 22 (2 marks)

Criteria	Marks
• Provides correct answer	2
• Makes progress towards answer	1

Sample answer:

$$A = \frac{85}{2}(160 + 180) + \frac{85}{2}(180 + 140)$$

$$A = 28050\ m^2$$

Question 23 (3 marks)

Criteria	Marks
• Provides correct answer	3
• Makes progress towards answer	2
• Provides some relevant information	1

Sample answer:

$reaction\ time\ distance = \frac{60 \times 1000}{3600} \times 1.5$
= 25 m
$braking\ distance = 0 \cdot 01(60)^2$
= 36 m
Total stopping distance: 25 + 36 = 61 m

Question 24 (5 marks)

a

Criteria	Marks
• Provides correct answer	1

Sample answer:

$C = 3000 + 45(75)$

$C = \$6375$

b

Criteria	Marks
• Provides correct answer	2
• Makes progress towards answer	1

Sample answer:

$8535 = 3000 + 45n$

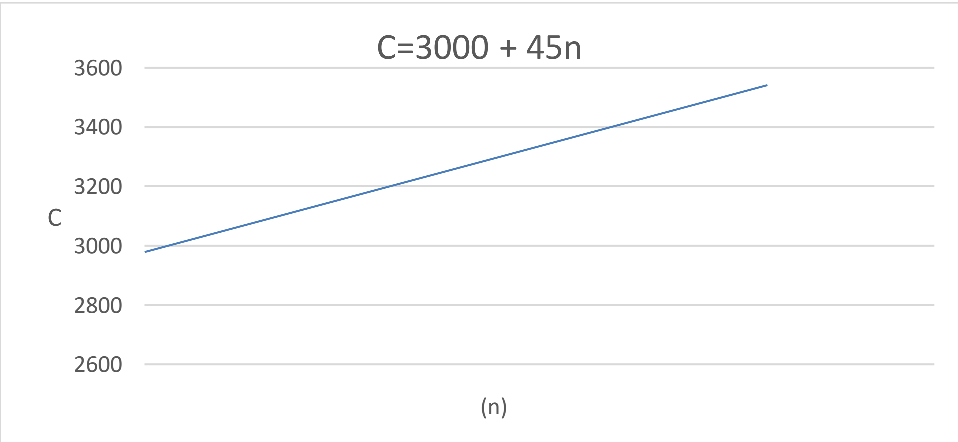
$45n = 5535$

$n = 123$

c

Criteria	Marks
• Provides correct answer	2
• Makes progress towards answer	1

Sample answer:



Question 25 (6 marks)

a

Criteria	Marks
• Provides correct answer	1

Sample answer:

$\frac{2.88}{24} = 0.12$ kwh/h

b

Criteria	Marks
• Provides correct answer	1

Sample answer:
5 weekdays in a week
5 hours/day peak
Over 4 weeks

0.12 x 5 x 5 x 4
= 12 kWh

c

Criteria	Marks
• Provides correct answer	2
• Makes progress towards answer	1

Sample answer:
19 hours/day off-peak over 5 weekdays
24 hours/day off-peak over2 weekends
Over 4 weeks

[(0.12 x 19 x 5) + (0.12 x 24 x 2)] x 4
= 68.64 kWh

d

Criteria	Marks
• Provides correct answer	2
• Makes progress towards answer	1

Sample answer:
Peak: 12 x 0.34 = $4.08
Off-peak: 68.64 x 0.16 = $10.98

Total cost: 4.08 + 10.98
= $15.06

Question 26 (9 marks)

a

Criteria	Marks
• Provides correct answer	1

Sample answer:

Median lies between the scores of 22 and 29

$$\frac{22+29}{2} = 25.5$$

Median = 25.5

b

Criteria	Marks
• Provides correct answer	2
• Makes progress towards answer	1

Sample answer:

11, 12, 12, 13, 22, 29, 31, 33, 34, 49

$Q_1 = 12$

$Q_3 = 33$

$IQR = 33 - 12$

$= 21$

c

Criteria	Marks
• Provides correct answer	3
• Makes progress towards answer	2
• Provides some relevant information	1

Sample answer:

Less than $Q_1 - 1.5 \times IQR$

or

More than $Q_3 + 1.5 \times IQR$

$\text{Less than } 12 - 1.5 \times 21$

< -19.5

$\text{More than } 33 + 1.5 \times 21$

> 64.5

$\text{Outliers are } < -19.5 \text{ or } > 64.5$

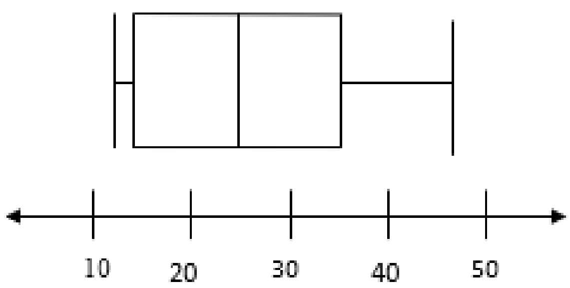
$\therefore$ There are no outliers in this data

d

Criteria	Marks
• Provides correct answer	3
• Makes progress towards answer	2
• Provides some relevant information	1

Sample answer:

$Min = 11$
$Q_1 = 12$
$Median = 25.5$
$Q_3 = 33$
$Max = 49$

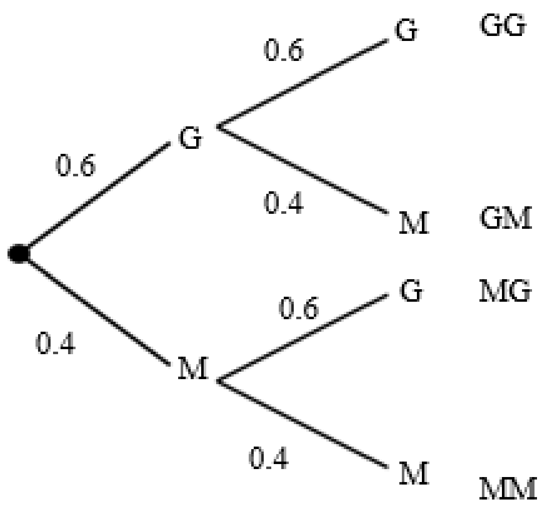


Question 27 (5 marks)

a

Criteria	Marks
• Provides correct answer	2
• Makes progress towards answer	1

Sample answer:



b

Criteria	Marks
• Provides correct answer	2
• Makes progress towards answer	1

Sample answer:

GM or MG: (0.6 x 0.4) + (0.4 x 0.6)
P = 0.48

c

Criteria	Marks
• Provides correct answer	1

Sample answer:

0.6 x 20 = 12 goals

Question 28 (5 marks)

a

Criteria	Marks
• Provides correct answer	3
• Makes progress towards answer	2
• Provides some relevant information	1

Sample answer:

Cost of car: $32 500
Stamp duty: 3% x 32 500 = $975
CTP Insurance: $31.50 x 12 = $378
Registration: 212 x 2 = $424
Total cost: 32500 + 975 + 378 + 424 = $34 277

b

Criteria	Marks
• Provides correct answer	2
• Makes progress towards answer	1

Sample answer:

$S = V_0 - Dn$
$V_0 = 32\ 500$
$D = 0.08 \times 32\ 500$
$\quad = 2600$
$n = 4$
$S = 32\ 500 - (2600 \times 4)$
$S = \$22\ 100$

Question 29 (3 marks)

Criteria	Marks
• Provides correct answer	3
• Makes progress towards answer	2
• Provides some relevant information	1

Sample answer:

$Taxable\ income = Earnings - Deductions$

$= 72\,816 - 520 - 212\ (Petrol\ for\ personal\ use\ is\ not\ a\ deductable\ expense)$

$= 72\,084$

$Tax\ payable = 5\,092 + 0.325 \times (72\,084 - 45\,000)$

$\$13\,894.30$

Question 30 (6 marks)

a

Criteria	Marks
• Provides correct answer	1

Sample answer:

Precision = 1

b

Criteria	Marks
• Provides correct answer	1

Sample answer:

Absolute error = 0.5 x precision

0.5 x 1 = $\pm$ 0.5

c

Criteria	Marks
• Provides correct answer	1

Sample answer:

$Percentage\ error = \frac{Absolute\ error}{Measurement} \times 100$

$\frac{0.5}{29} \times 100$

= 1.7%

d

Criteria	Marks
• Provides correct answer	3
• Makes progress towards answer	2
• Provides some relevant information	1

Sample answer:
Upper bound: 29.5cm by 38.5 cm
Lower bound 28.5 cm by 37.5 cm

Upper Perimeter: 29.5 + 29.5 + 38.5 + 38.5
= 136cm

Lower Perimeter: 28.5 + 28.5 + 37.5 + 37.5
= 132

$132 < P < 136$

Question 31 (4 marks)

Criteria	Marks
• Provides correct answer	4
• Makes significant progress towards answer	3
• Makes progress towards answer	2
• Provides some relevant information	1

Sample answer:
S.A = $2\left(\frac{\theta}{360}\pi r^2\right) + \frac{\theta}{360}2\pi rh + 2(lb)$

$$750 = 2\left(\frac{200}{360}\pi 5^2\right) + \frac{200}{360}2\pi(5)(x) + 2(5)(x)$$

$27.45x = 662.73$
$x = 24.14\ cm$
$x = 241\ mm$

2022 Mathematics Standard 2 Year 11
Mapping Grid

Section I

Question	Marks	Syllabus outcomes
1	1	M2
2	1	S1.1
3	1	M.1
4	1	F1.3
5	1	S2
6	1	A2
7	1	A1
8	1	M1.3
9	1	F1.3
10	1	M1.1

Section II

Question	Marks	Syllabus outcomes
11	1	S1.1
12	1	M1.1
13	1	M1.3
14	2	F1.3
15	2	A1
16	2	M1.2
17	3	A2
18	2	M1.3
19	2	A2
20	3	M1.2
21	3	F1.2
22	2	F1.1
23	3	A2
24	5	A1
25	6	M1.1
26	9	S2
27	5	S1.1
28	5	F1.2
29	3	F1.3
30	6	M1.1
31	4	M1.3, A1